Customer Shopping Behaviour Analysis

1. Project Overview

This project analyses customer shopping behaviour using transactional data of **3,900 records**. The objective is to identify customer purchasing patterns, product preferences, and business trends to support data-driven decision-making.

The analysis focuses on understanding how factors like **age, gender, subscription status, discounts, and purchase history** influence customer behaviour.

2. Dataset Summary

- **Total Rows:** 3,900
- **Total Columns:** 18

Key Features:

- **Customer Details:** Age, Gender, Location
- **Purchase Information:** Item Purchased, Category, Purchase Amount
- **Behaviour Data:** Previous Purchases, Frequency of Purchases
- **Business Factors:** Discount Applied, Promo Code Used, Shipping Type
- **Customer Feedback:** Review Rating

Missing Data:

- Review Rating column had missing values (handled during pre-processing)

3. Data Pre-processing using Python

Data cleaning and preparation were performed using Python (Pandas):

Steps Performed:

- Imported dataset using pandas
- Checked structure using df.info()
- Generated summary statistics using df.describe()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Paym Meth
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ven
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	C
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Cre C
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Pay
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Pay

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Handling Missing Values

- Missing values in **Review Rating** were filled using:
  - Median value grouped by category

```
[9]: df["Review Rating"] = df.groupby("Category")["Review Rating"].transform(lambda x: x.fillna(x.median()))
```

```
[10]: df.isnull().sum()
```

```
[10]: Customer ID               0
      Age                       0
      Gender                    0
      Item Purchased            0
      Category                  0
      Purchase Amount (USD)     0
      Location                  0
      Size                      0
      Color                     0
      Season                    0
      Review Rating             0
      Subscription Status       0
      Shipping Type             0
      Discount Applied          0
      Promo Code Used           0
      Previous Purchases        0
      Payment Method            0
      Frequency of Purchases    0
      dtype: int64
```

5. Column Standardization

- Renamed columns into **snake_case format**
  Example:
- Customer ID → customer_id
- Purchase Amount → purchase_amount

```
[11]: df.columns = df.columns.str.lower().str.replace(' ', '_').str.replace('[^a-z0-9_]', '', regex=True)
      df=df.rename(columns={"purchase_amount_usd":"purchase_amount"})
```

```
[12]: df.columns
```

```
[12]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
             'purchase_amount', 'location', 'size', 'color', 'season',
             'review_rating', 'subscription_status', 'shipping_type',
             'discount_applied', 'promo_code_used', 'previous_purchases',
             'payment_method', 'frequency_of_purchases'],
            dtype='object')
```

Final Dataset

- Cleaned dataset saved as:

processed_customer_shopping_behavior.cs

4. Data Analysis using SQL

After cleaning, data was analyzed using SQL to answer business questions.

1. Revenue by Gender

- Compared total revenue generated by male and female customers

Result Grid | Filter Row

gender	revenue
Male	89125
Female	41897

2. High-Spending Discount Users

- Identified customers who used discounts but still spent above average

Result Grid | Filter Rows:

customer_id	purchase_amount
2	64
3	73
4	90
9	97
12	68
13	72
16	81
24	88

customer_data 3

3. Top 5 Products by Rating

- Identified highest-rated products

Result Grid | Filter Rows: | Export:

item_purchased	Average Producr Rating
Boots	3.9535211267605623
Gloves	3.9268292682926824
Sandals	3.832631578947369
T-shirt	3.8274999999999992
Shorts	3.8093023255813945

Result 4

Output

4. Shipping Type Comparison

- Compared average purchase for:
  - Standard vs Express

Result Grid | Filter Rows:

shipping_type	avg_purchase_amount
Express	61.33
Standard	57.76

5. Subscribers vs Non-Subscribers

- Compared:
  - Total customers
  - Average spend
  - Total revenue

Result Grid | Filter Rows: | Export: | Wrap Cel

subscription_status	total_customer	avg_spend	total_revenue
No	1587	59.6648	94688
Yes	619	58.6979	36334

6. Discount-Dependent Products

- Identified products highly dependent on discounts

Result Grid | Filter Rows:

item_purchased	discount_rate
Sneakers	56.67
Sweater	53.76
Coat	51.04
Jewelry	48.35
Hat	47.73

7. Customer Segmentation

Customers classified as:

- New
- Returning
- Loyal

Result Grid | Filter Rows:

customer_segment	total_customers
Loyal	1973
Returning	233

8. Top 3 Products per Category

- Identified best-selling items in each category

Result Grid | Filter Rows: | Exp

category	item_purchased	total_purchases
Accessories	Scarf	96
Accessories	Jewelry	91
Accessories	Hat	88
Accessories	Belt	88
Clothing	Shirt	104
Clothing	Dress	101
Clothing	Pants	99
Footwear	Sandals	95

10. Revenue by Age Group

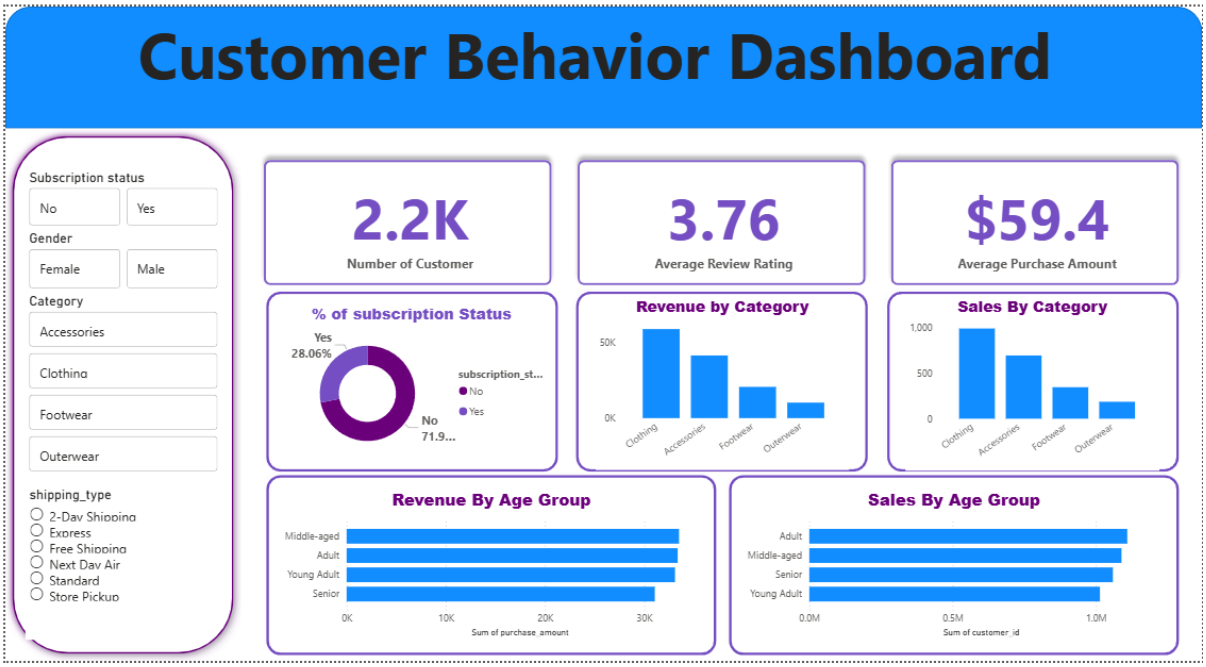
- Calculated revenue contribution by age group

Result Grid | Filter Rows:

age_group	total_revenue
Middle-aged	33502
Adult	33366
Young Adult	33095
Senior	31059

5. Dashboard in Power BI

An interactive dashboard was created in Power BI to visualize insights.



Dashboard Includes:

- Total Customers
- Average Purchase Amount
- Average Review Rating
- Revenue by Category
- Revenue by Age Group
- Subscription Distribution

6. Business Recommendations

- **Increase Subscriptions**
  Offer exclusive deals and benefits
- **Focus on Loyal Customers**
  Provide rewards and loyalty programs
- **Optimize Discount Strategy**
  Avoid over-dependency on discounts
- **Promote Top Products**
  Highlight best-selling and highly rated items
- **Target Key Age Groups**
  Focus marketing on high-revenue segments

7. Conclusion

This project successfully analyzed customer shopping behavior and identified key patterns in purchasing trends. The insights can help businesses improve marketing strategies, optimize pricing, and enhance customer retention.